

Causal Inference Crash Course:

Regression Discontinuity Design

Julian Hsu

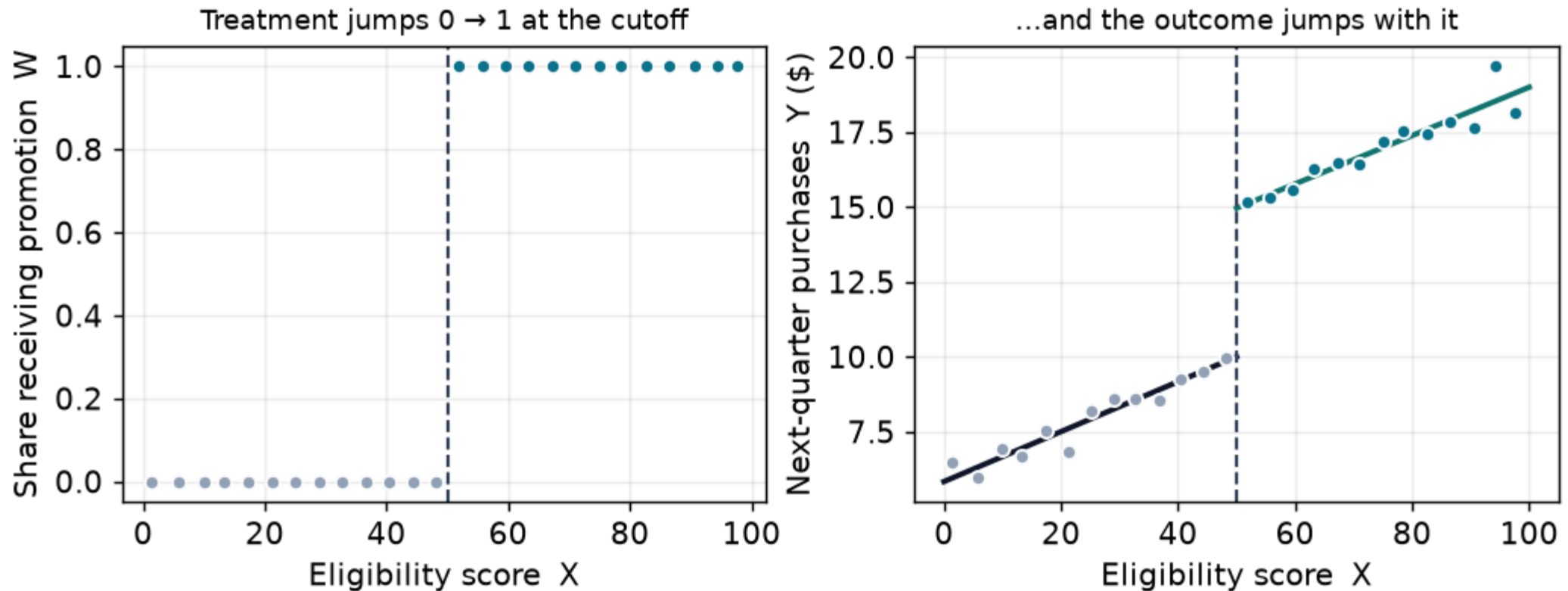
Overview

- **Regression discontinuity (RDD)**: a local natural experiment hiding at a cutoff.
- Earlier decks controlled for confounders (unconfoundedness). RDD instead assumes only **continuity** at the cutoff.
- We cover:
 - The big idea and the *local* estimand
 - Sharp vs. fuzzy designs
 - Estimation: one regression, plus three practical questions
 - Validating the design
 - A simulated worked example
- Assumes the earlier decks: potential outcomes, OLS, inference.

A threshold-based policy

- A **promotion** (W_i) turns on once an **eligibility score** (X_i) crosses a cutoff c .
 - e.g. a spend threshold for free shipping, a credit-score cutoff for an offer.
- We want its effect on **next-quarter purchases** (Y_i).
- Customers just below and just above c are nearly identical — only the promotion differs.
- That threshold is the experiment.

The Big Idea



- At the cutoff, treatment switches on (left) — and the outcome jumps with it (right).
- The jump in Y is the treatment effect.

Why not just run a regression?

- The simple linear model $Y_i = \alpha + \tau W_i + \beta X_i + \epsilon_i$ uses **all** the data and assumes one straight line.
- But X_i *perfectly determines* W_i : no overlap, so τ leans entirely on that functional form.
- RDD trusts the comparison **only at the cutoff**:

$$\tau_{RD} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x]$$

The assumption: continuity

- $E[Y_i(0) \mid X]$ and $E[Y_i(1) \mid X]$ do not jump at c .
- So absent treatment, nothing else changes at the cutoff — only W_i can make Y jump.
- Unlike unconfoundedness, we needn't observe confounders — only that they move **smoothly** through c .
- The biggest threat: **manipulation** of X_i to land just past c .
- Also audit for a **shared cutoff**: if another program keys on the same threshold, the jump bundles both effects.

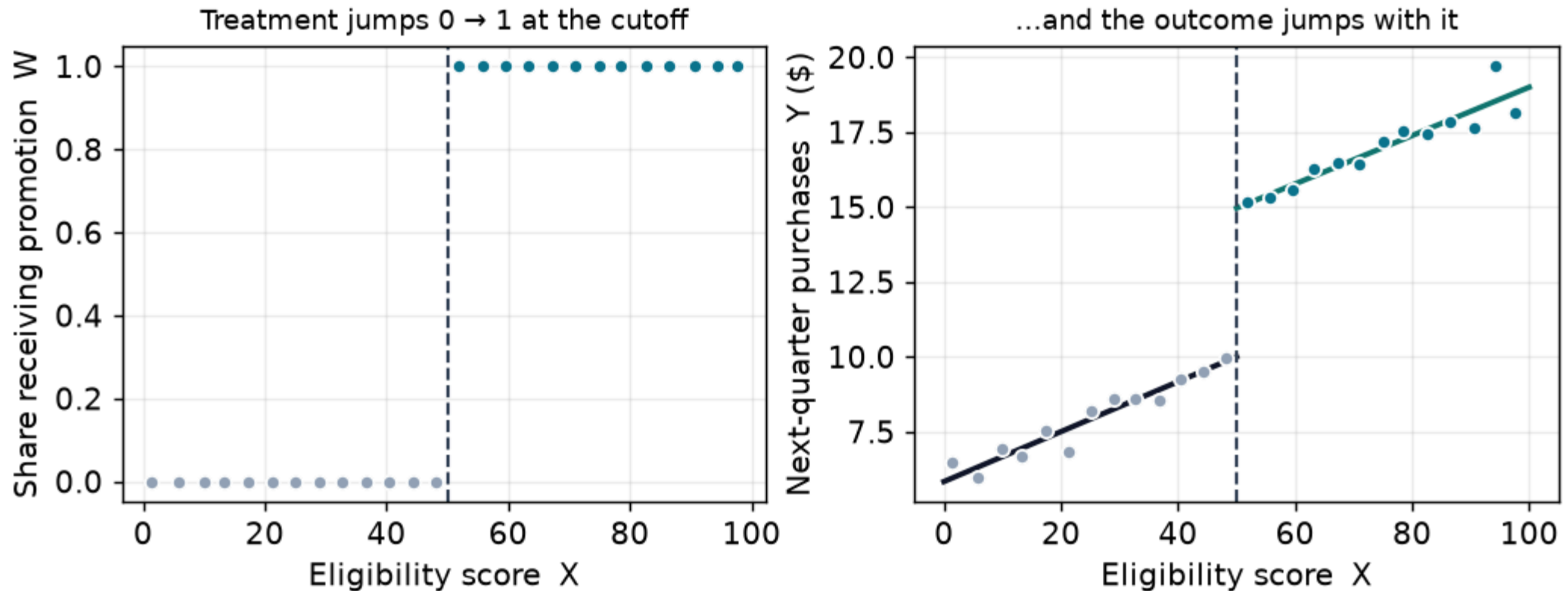
RDD estimates a *local* effect

- τ_{RD} is the effect **at the cutoff**, for units with $X_i \approx c$.
- Credible internally; limited externally — it needn't hold far from c .
- So you need data near c , and your answer is *about* c .
- The flip side: when the decision *is* the threshold — "should we move the cutoff?" — the effect at c is exactly the number you need.

Sharp vs. Fuzzy

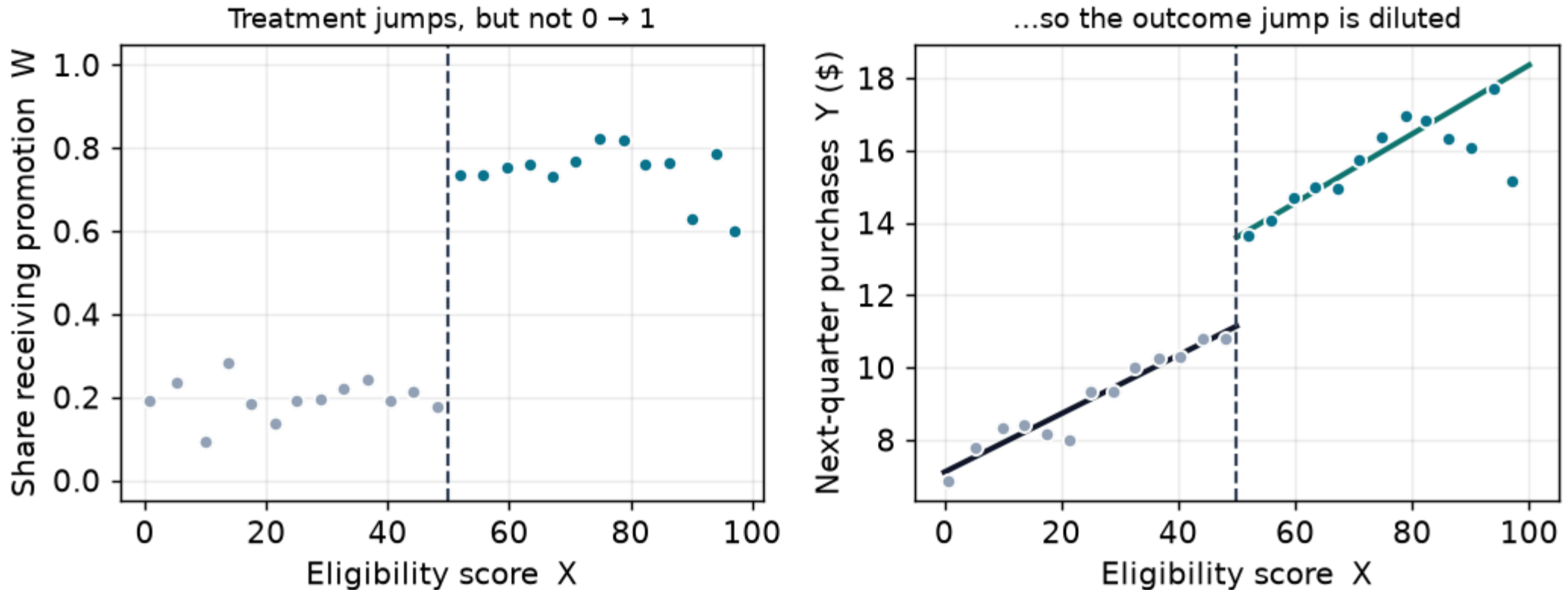
Does treatment switch on completely at the cutoff, or only partly?

Sharp: treatment jumps 0 → 1



- Everyone above the cutoff is treated, no one below: $W_i = 1\{X_i \geq c\}$.
- The jump in Y is the treatment effect, directly.

Fuzzy: treatment jumps only partway



- Same two pictures — but now crossing c only *raises the chance* of treatment (some below get it, some above don't).
- The outcome jump is **diluted**: only the extra treated units contribute.

Fuzzy RD: rescale the diluted jump

- Divide the outcome jump by the treatment jump — both read straight off the two panels:

$$\tau_{FRD} = \frac{\text{jump in } E[Y \mid X] \text{ at } c}{\text{jump in } E[W \mid X] \text{ at } c}$$

- This is IV, with $1\{X_i \geq c\}$ as the instrument — a LATE for compliers at the cutoff.
- Being IV, it carries the usual extra assumptions: **exclusion** (crossing c moves Y only through W) and **monotonicity** (crossing c pushes no one *out* of treatment).
- Sharp is the special case where the denominator is 1.
- Estimation and a worked demo: [Appendix — fuzzy RD in practice](#).

Estimation

One OLS regression, near the cutoff.

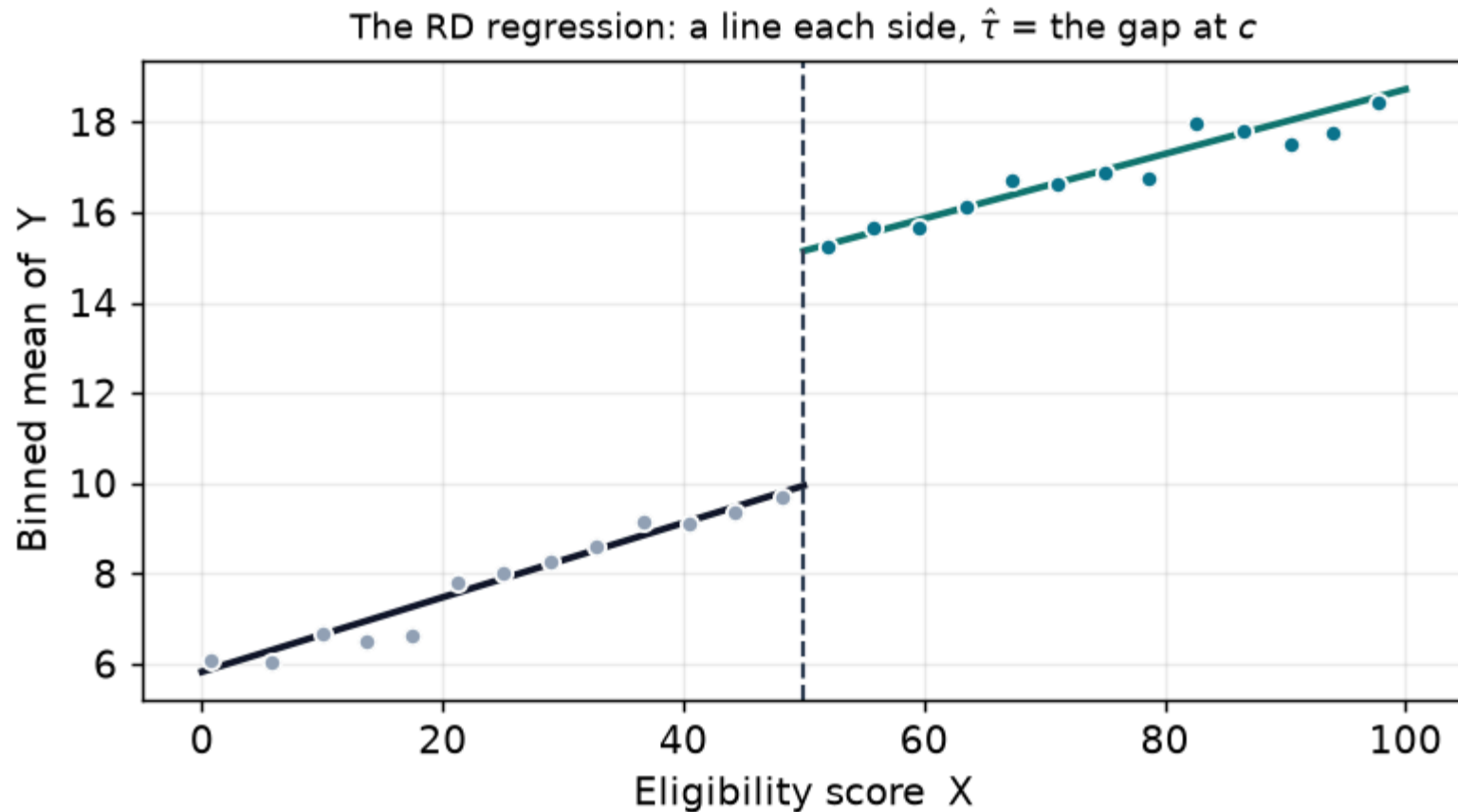
The RD regression

- Center the score at the cutoff and run **one OLS**, fully interacting treatment with the running variable:

$$Y_i = \alpha + \tau W_i + \beta(X_i - c) + \gamma W_i(X_i - c) + \epsilon_i$$

- Each piece has a job:
 - β : the slope below the cutoff
 - $\beta + \gamma$: the slope above — the interaction lets the sides differ
 - τ : the gap between the two lines at $X_i = c$ — the treatment effect
- Fit it using only observations near the cutoff ($|X_i - c| \leq h$).
 - (In practice the window is usually kernel-weighted too — [Appendix — kernel weights and RBC detail.](#))

What it looks like



- One regression, drawn: a line on each side, and $\hat{\tau}$ is the gap at the cutoff.

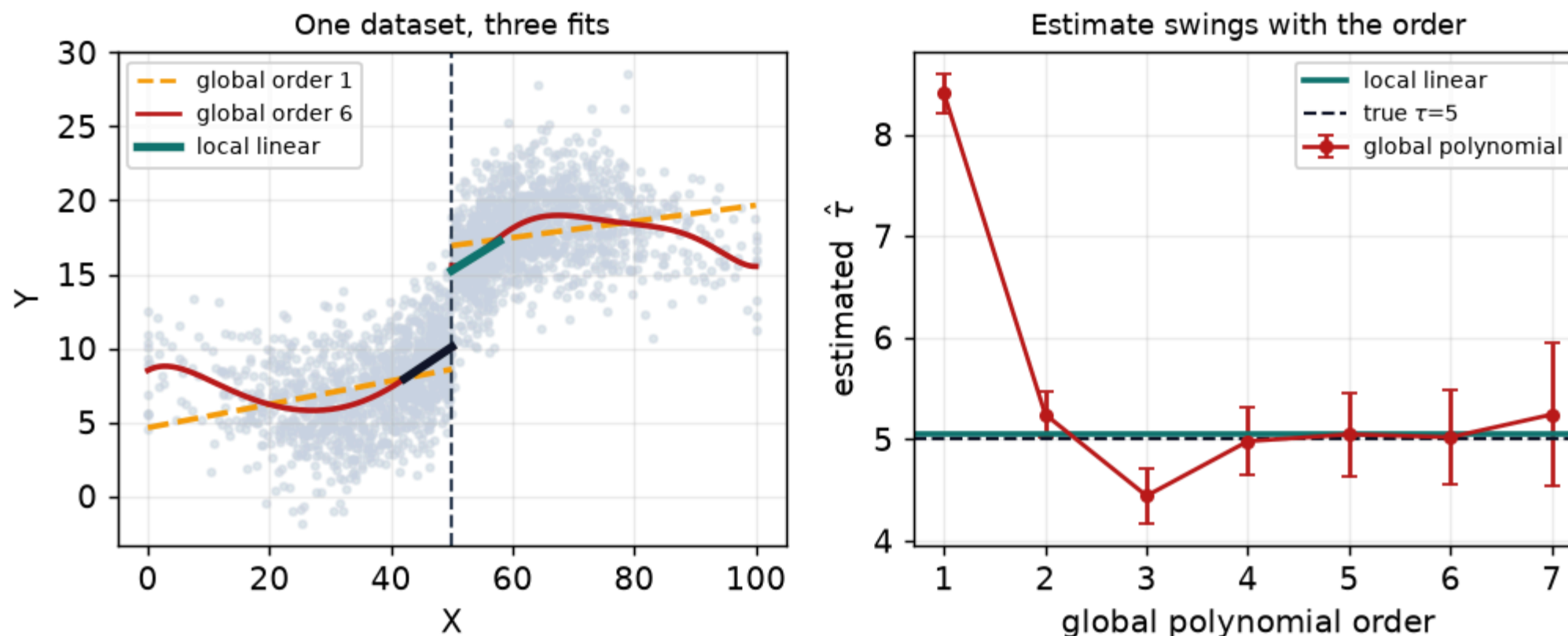
Question 1: what if the relationship is curved?

- If $E[Y \mid X]$ bends inside the window, straight lines mis-measure the jump.
- Natural fix: add polynomial terms, still interacted with W_i :

$$Y_i = \alpha + \tau W_i + \sum_{k=1}^p [\beta_k (X_i - c)^k + \gamma_k W_i (X_i - c)^k] + \epsilon_i$$

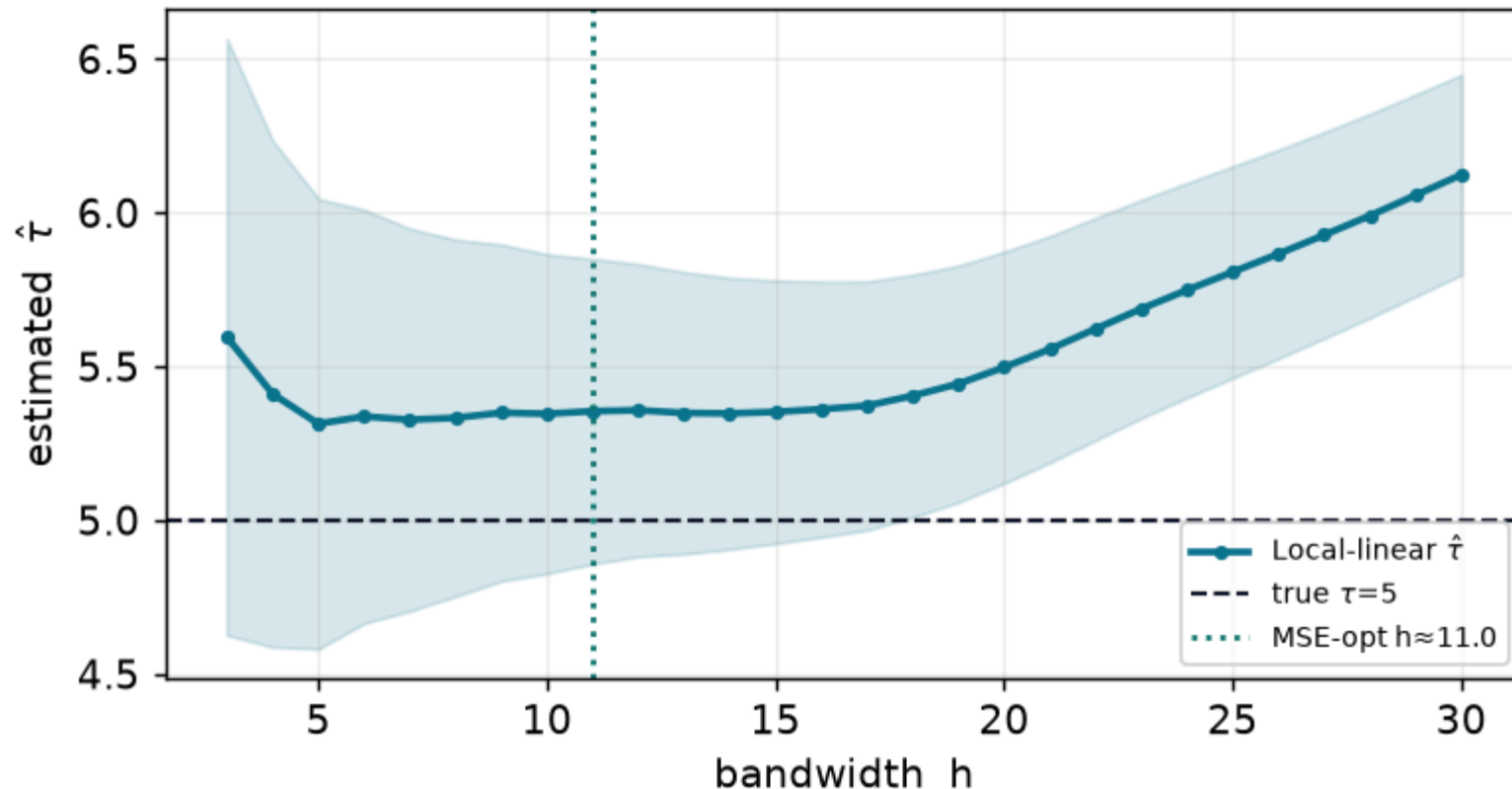
- Keep p **low** (1 or 2). Higher orders chase noise — next slide.

Why not a high-order polynomial on all the data?



- Far-away data then controls the fit at the cutoff, and $\hat{\tau}$ **swings with the order** (Gelman & Imbens 2019).
- Curvature is better handled by *narrowing the window* than by raising the order.

Question 2: how wide a window? Bias vs. variance



- **Narrow h :** a straight line is locally right (*low bias*), but few observations (*high variance*).
- **Wide h :** more observations (*low variance*), but the line is more wrong (*high bias*).

- The **MSE-optimal** bandwidth (CCT 2014) balances the two — a data-driven answer, not a hand-picked one.

Question 3: can I use the usual confidence intervals?

- No — and the reason is the bandwidth choice you just made.
- The MSE-optimal h *accepts some bias on purpose* (that's the trade-off). So $\hat{\tau}$ is centered slightly off the truth.
- A textbook CI is built around $\hat{\tau}$ — centered on a biased point, it covers the truth **less** than 95% of the time.
- **Robust bias-corrected (RBC)** inference: estimate the bias, subtract it, and widen the CI for the extra noise of estimating it. That is `rdrobust`'s **Robust** row.

Validating the design

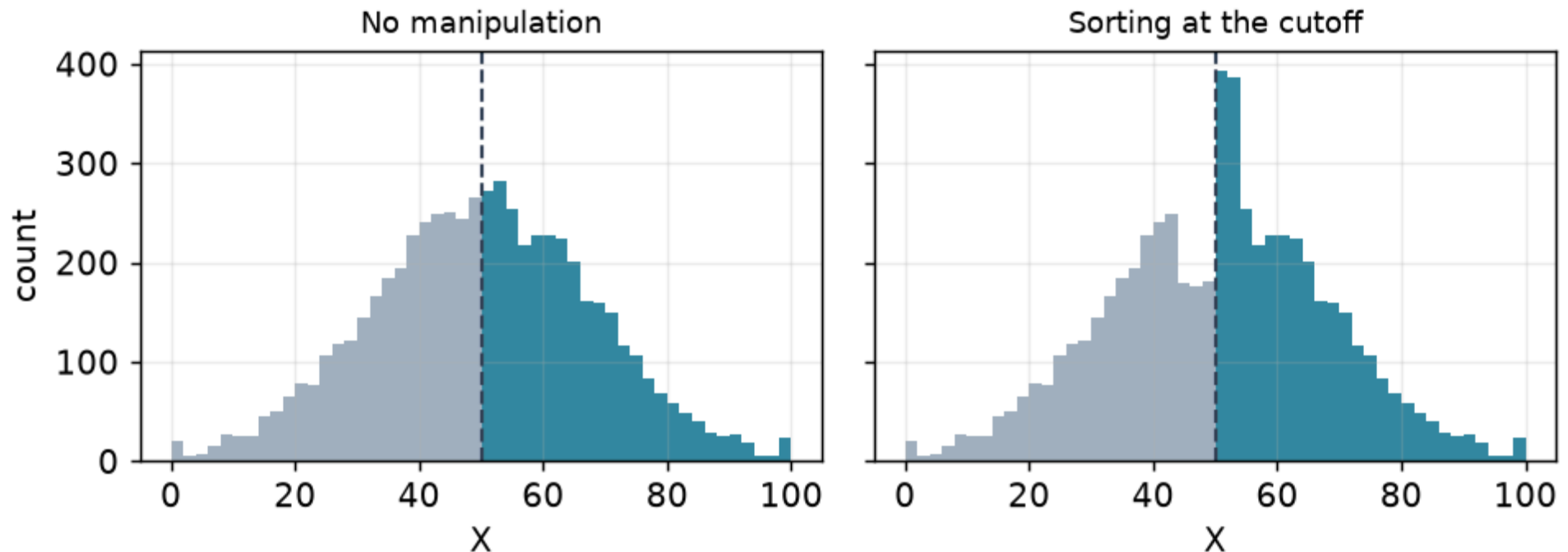
Continuity has testable implications.

Continuity is testable

- If continuity holds:
 1. the density of X_i doesn't jump at c (no manipulation);
 2. predetermined covariates don't jump at c ;
 3. fake cutoffs show no effect;
 4. the estimate survives dropping points at the cutoff (donut) and changing h .
- These are tests of *implications*: failing one refutes the design, but passing all four earns credibility, not proof.

Test 1: no manipulation

- If customers can nudge their score, the ones just above c chose to be there — not comparable to those just below.
- The tell is a **pile-up** just above c :

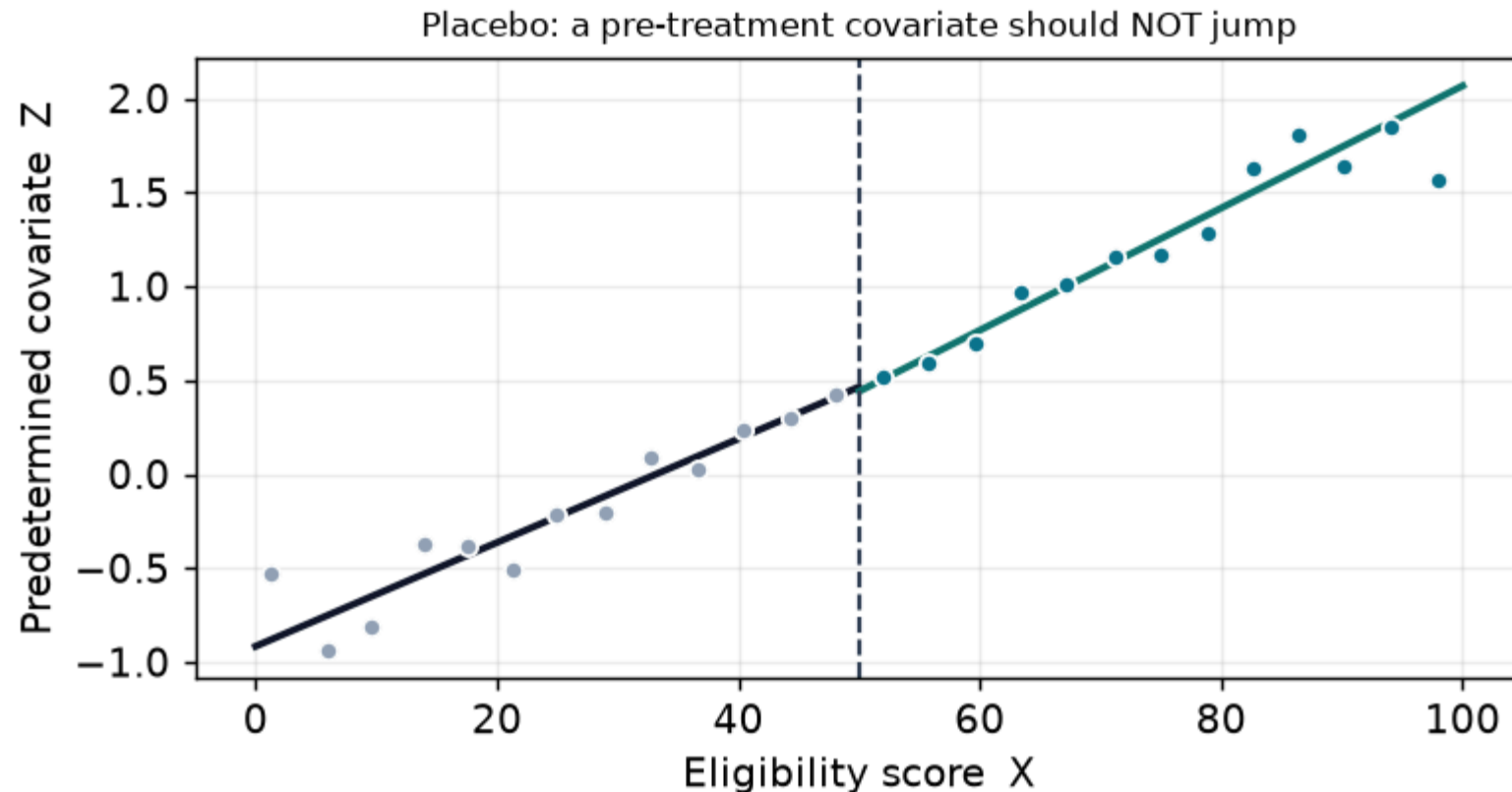


- Formal test: is the density of X_i continuous at c ? (McCrary 2008; `rddensity`.)

- Nuance: customers may *influence* their score — that alone is fine. The design breaks only under **precise** control over landing just past c (Lee 2008).

Test 2: covariates don't jump

- Just-left and just-right customers should look identical *before* treatment.
- Check it directly: rerun the **same RD regression** with a pre-treatment covariate as the outcome. The "effect" should be zero.



Test 3: placebo cutoffs

- Move the cutoff to a value where nothing actually changes — say $c = 40$ or $c = 60$ — and re-run the RD.
- No policy switches there, so the true jump is zero.
- Fit each placebo on one side of the real cutoff only (e.g. only $X_i < 50$ data for $c = 40$), so the true jump at 50 can't contaminate it.
- If you still find a "jump", your specification finds jumps everywhere, and the one at c means nothing.

Test 4: the donut

- Manipulators land **right at the cutoff** — the observations closest to c are the least trustworthy.
- The donut test: drop points within $\pm r$ of c , refit, and widen the hole. A clean design barely moves; a sorted one moves a lot.
- Worked example with figures: [Appendix — the donut test](#).

Test 5: bandwidth sensitivity

- Re-estimate across a range of bandwidths (the bias–variance figure already shows this curve).
- A credible estimate is **stable** over reasonable h — not a knife-edge result that appears at exactly one bandwidth.

Worked example

Simulated data, so we know the answer ($\tau = 5$).

Two scenarios

- Score $X_i \in [0, 100]$, cutoff $c = 50$, promotion $W_i = 1\{X_i \geq c\}$, outcome Y_i = next-quarter purchases, true $\tau = 5$.
1. **Clean** — continuity holds and $E[Y \mid X]$ is a straight line.
 2. **Nonlinear** — continuity still holds, but $E[Y \mid X]$ is curved. Stress-tests the functional form.
- What if continuity *fails*? See [Appendix — the donut test](#).
 - Companion notebook with all code [here](#).

Four models

1. **Naive difference**: mean Y above the cutoff minus mean Y below. Ignores X_i entirely.
2. **Global polynomial**: the RD regression on *all* the data, with 4th-order $(X_i - c)$ terms, interacted.
3. **Local linear**: the RD regression from the estimation section — linear, within $h = 8$ of the cutoff, triangular weights.
4. **rdrobust**: local linear with the MSE-optimal bandwidth and robust bias-corrected confidence intervals.

Bias by model and scenario

- Mean bias $(\hat{\tau} - \tau)$ over 500 simulations; $\tau = 5$.

Model	Clean	Nonlinear
Naive difference	+2.28	+5.33
Global polynomial	+0.01	-0.04
Local linear	+0.02	+0.05
<code>rdrobust</code> (RBC)	+0.00	-0.06

- The naive difference is biased even in the clean case — it compares high scorers to low scorers, not the two sides of the cutoff.
- All three RD models handle the curvature.

Coverage

- How often the 95% confidence interval contains the truth:

Model	Clean	Nonlinear
Local linear (naive SE)	0.95	0.94
<code>rdrobust</code> (RBC)	0.94	0.95

- With a valid design, the intervals deliver their nominal 95%.
- When the design is broken, they don't — see [Appendix — the donut test](#).

Conclusion

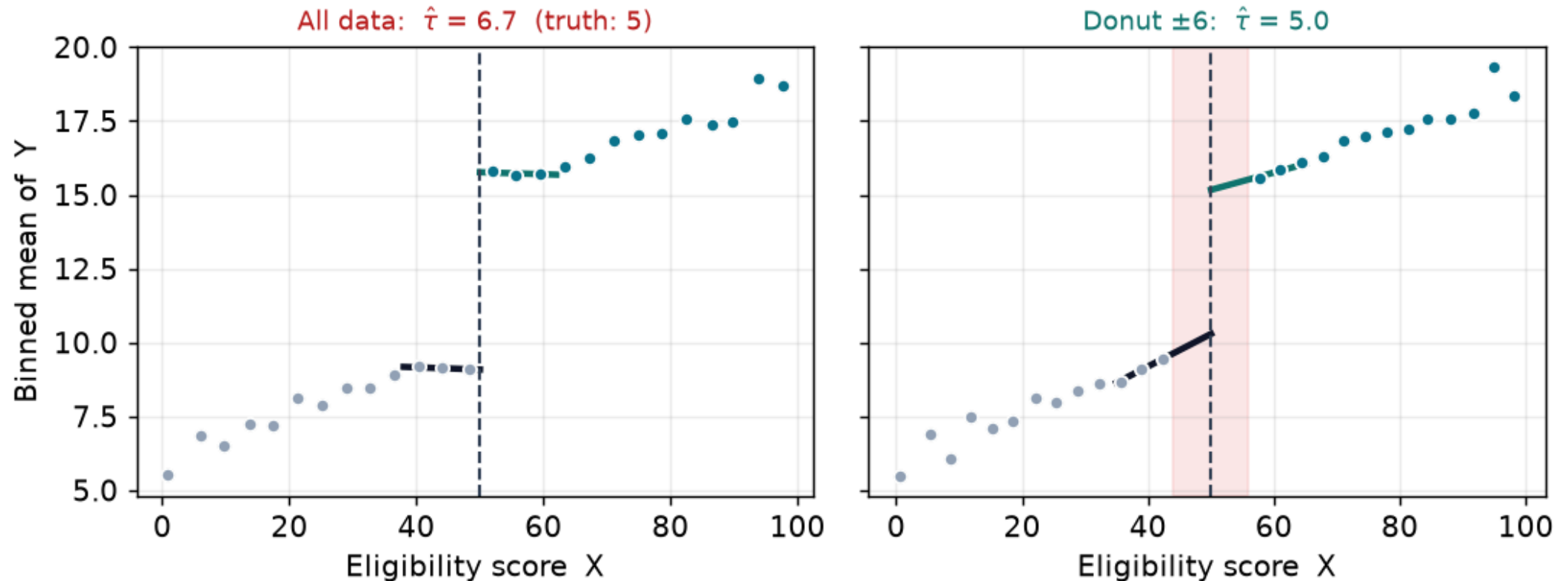
- RDD is a credible **local experiment** at a cutoff.
- Pros: transparent, a *testable* assumption, strong visuals.
- Cons: only a **local** effect, needs data near c , and it dies under manipulation.
- Reach for it when a real decision turns on a threshold — then run the checks.

Appendix Slides

Appendix: the donut test

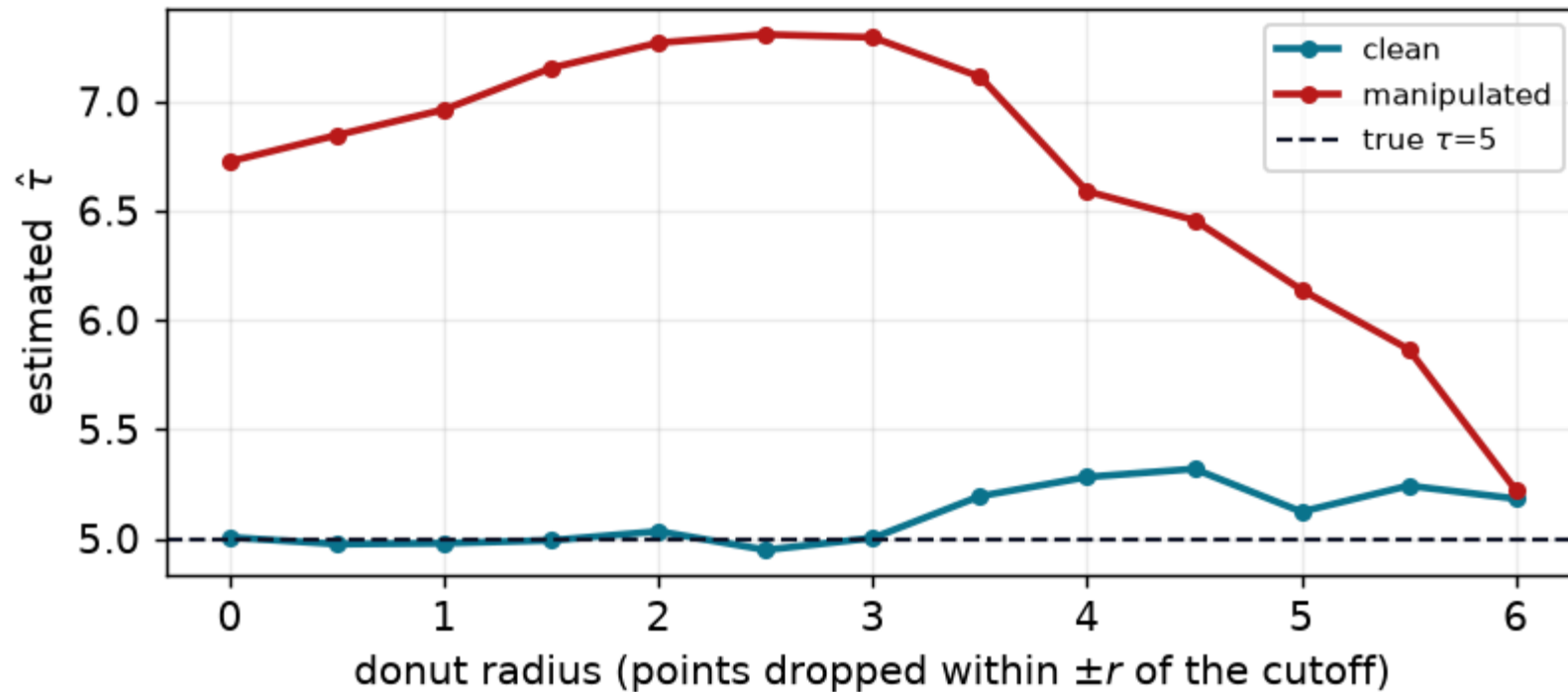
- The scenario, in our retail example: customers just below the cutoff nudge their score above it — retry a transaction, time a purchase — and the ones who bother are the **higher-spending** ones.
- The just-above group is no longer comparable to the just-below group: **continuity is broken**, and every estimator inherits the bias.
- The sorters sit **right at the cutoff**. The donut test exploits that: drop the points nearest c and see whether the estimate survives.

The donut in action



- Left: sorting drags the fit up — $\hat{\tau} = 6.7$ vs. a truth of 5. Note the tell: a dip just below the cutoff, a bulge just above.
- Right: drop the shaded ± 6 hole and refit from the outside — $\hat{\tau} = 5.0$.

Widening the donut



- Repeat over radii. Clean design: flat. Sorted design: the estimate moves a lot, drifting toward the truth as the sorters are dropped.
- The *movement* is the diagnostic — a stable path is evidence the design is clean.

No estimator survives manipulation

- The same 500-simulation study as the worked example, under the sorting scenario:

Model	Bias	95% CI coverage
Naive difference	+2.60	—
Global polynomial	+1.99	—
Local linear	+1.73	0.00
<code>rdrobust</code> (RBC)	+1.89	0.00

- Every model is biased and the intervals essentially *never* contain the truth.
- The donut **diagnoses** the problem; it does not license the original estimate. Validation comes first.

Appendix: fuzzy RD in practice

- In the field, eligibility rarely equals take-up: crossing c *unlocks* the promotion, but not everyone redeems it. Most industry threshold designs are fuzzy.
- Estimation is the same local machinery run twice — the jump in Y (intent-to-treat) divided by the jump in W (first stage). `rdrobust(y, x, fuzzy=w)` does both in one call.
- **Report the first stage.** A small jump in take-up is a weak instrument: intervals widen fast and the estimate turns fragile.
- Validation is unchanged — the density and covariate tests involve only X , so they don't care that treatment is fuzzy.
- Worked demo (hand-rolled Wald ratio vs. `rdrobust`) in the companion notebook.

Appendix: covariates and power

- **Covariates.** Predetermined covariates can't rescue a broken design; what they buy is *precision* (Calonico, Cattaneo, Farrell, Titiunik 2019; `rdrobust(covs=)`). In the companion notebook the robust CI narrows by about a quarter.
- If $\hat{\tau}$ moves when covariates enter, treat it as a red flag — smooth covariates shouldn't matter, so something is jumping at c .
- **Power.** Only observations near c carry information: in the notebook demo the RD interval is $\sim 3\times$ wider than an RCT on the *same* customers — roughly $10\times$ the sample for equal precision.
- Budget before you commit: `rdpower` does MDE and sample-size calculations for RD designs.

Appendix: cautions from the field

- **RD in time.** A launch date is not a cutoff: the same units sit on both sides, and trends, anticipation and autocorrelation — not continuity — drive the comparison (Hausman & Rapson 2018). Reach for event-study / DiD tools instead.
- **Extrapolation.** τ_{RD} is the effect at c . Costing a full rollout (or shutdown) means extrapolating away from the cutoff — that takes extra assumptions, and they should be stated, not smuggled.
- **Shared cutoffs.** Round numbers attract policies: if free shipping *and* a loyalty tier both switch at 50, *the jump price the bundle. Audit everything that changes at c .*

Appendix: the kink design (RKD)

- Sometimes treatment *intensity* is continuous at c but its **slope** changes (e.g. a benefit formula).
- RKD reads the effect from the change in **slope** of Y , not a jump in level. Same tools (`rdrobust`, `deriv=1`).

Appendix: extensions

- **Multiple / multi-dimensional cutoffs** — e.g. geographic RD.
- **Discrete running variables** — use the local randomization framework (Cattaneo, Idrobo, Titiunik 2024).
 - With few distinct score values, "close to c " is coarse: see also honest CIs (Kolesár & Rothe 2018) and `rdrobust`'s mass-point handling.
- **Local randomization** — treat a window around c as an experiment (`rdlocrand`).

Appendix: kernel weights and RBC detail

- The RD regression is usually fit by weighted least squares, down-weighting points far from the cutoff:

$$\min_{\alpha, \tau, \beta, \gamma} \sum_i K\left(\frac{X_i - c}{h}\right) (Y_i - \alpha - \tau W_i - \beta(X_i - c) - \gamma W_i(X_i - c))^2$$

- A triangular kernel K is MSE-optimal at a boundary; a uniform kernel = plain OLS in the window.
- RBC estimates the leading bias with a second (pilot) bandwidth $b \geq h$, subtracts it, and inflates the variance accordingly.

Literature Review of Related Papers

- **Foundational**

- Thistlethwaite & Campbell (1960). "Regression-Discontinuity Analysis." *J. Educational Psychology*.
- Hahn, Todd, Van der Klaauw (2001). "Identification and Estimation of Treatment Effects with a RD Design." *Econometrica*. [link](#)
- Lee (2008). "Randomized experiments from non-random selection in U.S. House elections." [link](#)
- Imbens & Lemieux (2008). "Regression discontinuity designs: A guide to practice." [link](#)
- Lee & Lemieux (2010). "Regression Discontinuity Designs in Economics." *J. Economic Literature*. [link](#)

- **Estimation & inference**

- Calonico, Cattaneo, Titiunik (2014). "Robust Nonparametric Confidence Intervals for RD Designs." *Econometrica*. [link](#)

- Calonico, Cattaneo, Farrell (2020). "Optimal bandwidth choice for robust bias-corrected inference in RD designs." [link](#)
- Calonico, Cattaneo, Farrell, Titiunik (2019). "Regression Discontinuity Designs Using Covariates." *REStat*. [link](#)
- Card, Lee, Pei, Weber (2015). "Inference on Causal Effects in a Generalized Regression Kink Design." *Econometrica*. [link](#)
- Kolesár & Rothe (2018). "Inference in RD Designs with a Discrete Running Variable." *AER*. [link](#)
- Gelman & Imbens (2019). "Why High-Order Polynomials Should Not Be Used in RD Designs." [link](#)

● Practical guides

- Cattaneo, Idrobo, Titiunik (2020, 2024). "A Practical Introduction to RD Designs: Foundations / Extensions." Cambridge. [link](#)
- Hausman & Rapson (2018). "Regression Discontinuity in Time: Considerations for Empirical Applications." *Annu. Rev. Resource Econ.* [link](#)

● Validation & software

- McCrary (2008). "Manipulation of the running variable in the RD design: A density test." [link](#)
- Cattaneo, Jansson, Ma (2020). "Simple Local Polynomial Density Estimators." (`rddensity`) [link](#)
- Cattaneo, Titiunik, Vazquez-Bare (2019). "Power Calculations for Regression-Discontinuity Designs." *Stata Journal*. (`rdpower`) [link](#)
- Software: `rdrobust`, `rddensity`, `rdlocrand`, `rdpower` — rdpackages.github.io

